Catalytic Quantum Error Correction: Theory, Efficient Catalyst Preparation, and Numerical Benchmarks

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We introduce Catalytic Quantum Error Correction (CQEC), a state recovery protocol exploiting catalytic covariant transformations [1]. CQEC recovers a known target state from noisy copies without an error *magnitude* threshold: recovery succeeds whenever the coherent modes satisfy $\mathcal{C}(\rho_0) \subseteq \mathcal{C}(\rho_{\text{noisy}})$, regardless of noise strength. The main practical bottleneck—catalyst preparation requiring $n^* \sim d^4 e^{2\gamma}$ copies—is resolved by a three-stage pipeline combining CPMG dynamical decoupling, Clifford twirling, and the recursive swap test, achieving $F_{\text{cat}} > 0.96$ with only 8 copies (10^9-fold reduction). Numerical validation across four quantum algorithms ($d = 4$–64), a cryptographic protocol, and three noise models confirms $F > 0.999$ in the asymptotic limit across 200 configurations.

I. INTRODUCTION

Quantum computers promise exponential speedups for problems in cryptography [2, 3], quantum simulation [4, 5], and machine learning [6, 7], but their practical realization is hindered by the fragility of quantum coherence under environmental decoherence [8, 9]. Quantum error correction (QEC) addresses this challenge by encoding logical information redundantly across physical qubits, enabling the detection and correction of errors below a code-specific threshold [10, 11]. However, the qubit overhead of conventional QEC is substantial: the Steane $[\![7, 1, 3]\!]$ code requires 7 physical qubits per logical qubit [12], while surface codes at distance d require $O(d^2)$ qubits [13, 14], and correction fails entirely when error rates exceed the threshold $p_{\text{th}} \approx 1\%$ [15].

A distinct approach to quantum state protection arises from the resource theory of quantum coherence [16, 17]. Coherence—the presence of off-diagonal elements in a state's density matrix with respect to a preferred basis—is a quantifiable resource under the framework of covariant (energy-conserving) operations [18, 19]. Shiraishi and Takagi [1] recently established that the transformation rate between coherent states can diverge arbitrarily: given n copies of a weakly coherent state ρ, one can produce $m \gg n$ copies of a target state ρ' via catalytic covariant operations, provided that the coherent modes of ρ' are contained within those of ρ. This result reveals that the output-to-input copy ratio m/n can be made arbitrarily large (the asymptotic rate diverges), creating an infinitely sharp boundary between recoverable and irrecoverable quantum states—a feature with no analog in conventional QEC.

The role of coherence in quantum thermodynamics and resource theories has been extensively studied [17–19], with catalytic transformations appearing in the context of thermodynamic state conversion [16] and entanglement manipulation. The interplay between symmetry constraints and quantum error correction has been studied in the context of covariant codes [20] and approximate QEC [21]. Covariant codes [20] encode information in a way that respects a symmetry group, with an inherent trade-off between code accuracy and the symmetry charge variance (the Eastin–Knill theorem for exact codes [22]). CQEC differs fundamentally: it does not encode information redundantly but instead uses catalytic transformations to amplify coherence in the original Hilbert space. However, the specific application of catalytic coherence amplification to quantum error correction has not been explored. In particular, it remains unclear whether the infinite amplification rate translates into effective protection for realistic quantum algorithms operating under decoherence, what the finite-copy resource costs are, and how these costs compare with conventional QEC.

Concurrently with this work, Raghoonanan and Byrnes [23] introduced Purification QEC (PQEC), which also uses N noisy copies and recursive swap tests to recover quantum states, but does *not* require knowledge of the target state. PQEC achieves doubly exponential purification ($\rho \mapsto \rho^N / \text{Tr}(\rho^N)$) with $O(M \log_2 N)$ qubits, and tolerates depolarizing errors up to $p = 3/4$. The relationship between CQEC and PQEC illuminates a fundamental trade-off: PQEC is state-agnostic but bounded by the noise threshold ($p_{\text{th}} = 3/4$ depolarizing, 1/2 dephasing), while CQEC exploits target-state knowledge to achieve *threshold-free* recovery at the cost of requiring this additional information. We provide a detailed comparison in Sec. VII C.

Moreover, even assuming the theoretical validity of catalytic recovery, the *practical* bottleneck is catalyst preparation. The constructive proof of Ref. [1] requires $n^* \sim d^4 e^{2\gamma}$ noisy copies to prepare a catalyst with fidelity $F_{\text{cat}} \geq 0.99$, where d is the Hilbert space dimension and γ is the dephasing strength. For a 6-qubit system ($d = 64$) under moderate dephasing ($\gamma = 2$), this translates to $n^* \approx 7.3 \times 10^{10}$ copies—far beyond current

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experimental capabilities. Recent advances in quantum state purification—including the recursive swap test of Childs *et al.* [24] (which we adopt in Sec. IV C) and related protocols [25, 26]—together with dynamical decoupling [27, 28] offer paths to dramatically better scaling.

In this paper, we bridge these gaps by (i) introducing the Catalytic Quantum Error Correction (CQEC) protocol with explicit success conditions, (ii) developing a quantum circuit architecture based on energy-conserving gates, (iii) proposing four catalyst preparation strategies culminating in a DD+Twirl+Swap Test pipeline that achieves a 10^9-fold copy-count reduction, and (iv) providing comprehensive numerical validation. Our central contributions are:

1. *Protocol and circuit.* We formulate the CQEC protocol with explicit success/failure conditions based on mode inclusion, implement it via energy-conserving gates in a minimal 4-qubit circuit, and provide finite-copy fidelity bounds.

2. *Efficient catalyst preparation.* We develop four catalyst preparation strategies culminating in the DD+Twirl+Swap Test pipeline, which achieves a 10^9-fold reduction in copy count over distillation under strong dephasing.

3. *Comprehensive benchmarks.* We validate CQEC across four quantum algorithms ($d = 4$–64), a cryptographic protocol, and three noise models (200 configurations), including comparison with conventional QEC codes, threshold verification across 10 orders of magnitude, and 100-cycle catalyst reuse.

While Theorems 1 and 2 are due to Shiraishi and Takagi [1], the application to quantum error correction, the circuit architecture, the catalyst preparation pipeline, and the comprehensive numerical validation are original to this work.

II. THEORETICAL FRAMEWORK

A. Unspeakable coherence and covariant operations

We work within the resource theory of unspeakable coherence [1, 18, 19, 29], where the free operations are covariant channels—completely positive trace-preserving maps Λ satisfying

$$\Lambda \circ \mathcal{U}_t = \mathcal{U}_t \circ \Lambda \quad \forall\, t \in \mathbb{R}, \tag{1}$$

where $\mathcal{U}_t(\rho) = e^{-iHt}\rho\, e^{iHt}$ is the time evolution generated by the system Hamiltonian $H = \sum_i E_i\, |i\rangle\langle i|$. Condition (1) enforces energy conservation: $[U, H_{\text{total}}] = 0$ for any unitary U implementing Λ [18]. This is physically motivated by the Wigner–Araki–Yanase theorem, which constrains measurements and operations that do not commute with conserved quantities [30].

For a d-level system, the ℓ_1-norm of coherence quantifies the total off-diagonal content:

$$C_{\ell_1}(\rho) = \sum_{i\neq j} |\rho_{ij}|. \tag{2}$$

The quantum Fisher information $\mathcal{F}(\rho, H) = 2\sum_{i,j} \frac{(\lambda_i-\lambda_j)^2}{\lambda_i+\lambda_j} |\langle i|H|j\rangle|^2$ [31] provides a complementary coherence monotone under covariant operations [1]. Unlike C_{ℓ_1}, QFI is operationally meaningful for metrology and satisfies $\mathcal{F}(\rho_{\text{noisy}}) \leq e^{-2\gamma}\mathcal{F}(\rho_0)$ under dephasing. We use QFI in Sec. VI B to quantify the coherence available for recovery. The two measures are complementary: C_{ℓ_1} counts total off-diagonal weight (relevant for mode coverage), while QFI weights contributions by their metrological utility. Under dephasing, both decay exponentially but at different rates: $C_{\ell_1} \sim e^{-\gamma}$ (slowest mode), $\mathcal{F} \sim e^{-2\gamma}$ (quadratic suppression), making QFI a more sensitive indicator of coherence loss.

B. Coherent modes and mode inclusion

The *modes of asymmetry* [19, 29] of a state ρ are defined as

$$\mathcal{D}(\rho) = \big\{\Delta_{ij} = E_i - E_j \mid \rho_{ij} \neq 0\big\}. \tag{3}$$

The *resonant coherent modes* $\mathcal{C}(\rho)$ are the integer-linear span of $\mathcal{D}(\rho)$:

$$\mathcal{C}(\rho) = \left\{ \sum_{(i,j)} n_{ij}\, \Delta_{ij} \;\middle|\; n_{ij} \in \mathbb{Z},\ \Delta_{ij} \in \mathcal{D}(\rho) \right\}. \tag{4}$$

$\mathcal{C}(\rho)$ forms a subgroup of $(\mathbb{R}, +)$ and determines the transformability of states under covariant operations [1].

Theorem 1 (Shiraishi–Takagi [1]). *If $\mathcal{C}(\rho') \subseteq \mathcal{C}(\rho)$ and ρ is full rank, the asymptotic marginal transformation rate*

$$R(\rho \to \rho') = \sup\left\{ \frac{m}{n} \;\middle|\; \rho^{\otimes n} \xrightarrow{\text{cov}} \sigma,\ \mathrm{Tr}_{[m]\setminus\{k\}}[\sigma] \approx \rho'\ \forall k \in [m] \right\}$$

diverges: $R(\rho \to \rho') = \infty$.

Theorem 2 (Shiraishi–Takagi [1]). *For any ρ, ρ' with $\mathcal{C}(\rho') \subseteq \mathcal{C}(\rho)$, there exists a catalyst c and a covariant operation Λ such that*

$$\mathrm{Tr}_C\big[\Lambda(\rho \otimes c)\big] = \rho', \quad \mathrm{Tr}_S\big[\Lambda(\rho \otimes c)\big] = c. \tag{5}$$

That is, $\rho \to \rho'$ is achievable by correlated-catalytic covariant transformation.

Crucially, for full-rank ρ, the mode inclusion condition is both necessary and sufficient: if $\mathcal{C}(\rho') \not\subseteq \mathcal{C}(\rho)$, then $R(\rho \to \rho') = 0$ and no catalytic transformation can achieve the conversion [1]. (The full-rank requirement is satisfied automatically for any finite noise channel applied to a pure state.) This provides a sharp classification of state pairs into recoverable ($R = \infty$) and irrecoverable ($R = 0$), with no intermediate regime.

We note that Theorem 2 establishes *correlated catalysis*: the output state $\tau = \Lambda(\rho \otimes c)$ satisfies $\mathrm{Tr}_C[\tau] = \rho'$ and $\mathrm{Tr}_S[\tau] = c$, but $\tau \neq \rho' \otimes c$ in general (the system and catalyst may be correlated in τ). This is weaker than *strict catalysis* ($\tau = \rho' \otimes c$), which imposes additional constraints [1]. For CQEC, correlated catalysis suffices: the catalyst is reusable because its reduced state is preserved, regardless of correlations with the recovered system. However, correlations between the catalyst and recovered system have operational consequences: (i) the system–catalyst correlations in τ mean that the recovered state $\rho' = \mathrm{Tr}_C[\tau]$ may differ from the state obtained by measuring the catalyst and post-selecting, and (ii) over multiple reuse cycles, correlations could accumulate in the joint state, potentially degrading effective catalyst quality. In our 100-cycle durability test (Sec. VI F), the catalyst deviation remains below 10^{-12} per cycle, indicating that correlation accumulation is negligible at this system size.

C. CQEC protocol

The CQEC protocol applies Theorem 2 to quantum state recovery. We use the term "recovery" rather than "correction" to emphasize the key operational distinction: CQEC recovers a *known* target state from noisy copies, whereas QEC corrects errors on an *unknown* encoded state. Despite the name "error correction," CQEC is more precisely a *catalytic state recovery* protocol. We retain "error correction" in the name because (i) the protocol corrects the effects of decoherence on quantum states, and (ii) the catalyst-mediated recovery is reusable across cycles, analogous to syndrome-based correction rounds. The requirement that ρ_0 be known does not reduce CQEC to trivial state re-preparation or tomography: the covariant channel Λ operates entirely in the quantum domain without measurement, preserving entanglement and coherence structure that tomography would destroy. The practical value lies in recovering expensive-to-prepare states cheaply from noisy copies rather than re-executing the original circuit. CQEC does not violate the no-cloning theorem [32] for three reasons: (i) the target state ρ_0 is *known*, so no unknown-state information is being duplicated; (ii) the n input copies are consumed (not preserved) by the protocol; and (iii) the covariant channel Λ is state-specific and cannot be applied to arbitrary unknown states.

Given an ideal state ρ_0 corrupted by decoherence to ρ_{noisy}, the protocol proceeds as follows.

Step 0 (Catalyst construction). Construct a full-rank catalyst c with $\mathcal{D}(c) \supseteq \mathcal{D}(\rho_0)$, following the construction of Proposition S.23 in Ref. [1]:

$$c = \frac{1}{n_{\text{cat}}} \sum_{k=1}^{n_{\text{cat}}} \rho^{\otimes(k-1)} \otimes \tau_{n_{\text{cat}}-k} \otimes |k\rangle\langle k|_R \,, \tag{6}$$

where n_{cat} is the number of catalytic branches, $\tau_{n_{\text{cat}}-k} = \mathrm{Tr}_{1,\ldots,k}[\sigma_{n_{\text{cat}}}]$ is the reduced state, and R is a register system of dimension n_{cat}. The apparent circularity is resolved by the constructive proof of Ref. [1]: two-level catalysts c_i are first constructed from μ_i copies via non-catalytic covariant operations, then composed into the full catalyst via Theorem S.10. The catalyst dimension is $\dim(c) = n_{\text{cat}} \cdot d^{n_{\text{cat}}-1} \cdot n_{\text{cat}}$, which grows exponentially. In our variational implementation (Sec. III C), we use a simplified catalyst of dimension $d_C = 2^{n_C}$ with $n_C = n_S$ qubits. This gap is fundamental: the variational circuit searches for a high-fidelity covariant transformation within a restricted ansatz, validating theoretical predictions but not implementing the full constructive proof. Efficient catalyst preparation strategies that dramatically reduce the required resources are developed in Sec. IV. Equation (6) defines the formal protocol; in practice, we replace it with the DD+Twirl+Swap Test pipeline (Sec. IV E) for catalyst preparation and the variational circuit (Sec. III C) for the recovery step.

Step 1 (Mode verification). Compute $\mathcal{D}(\rho_{\text{noisy}})$ and verify $\mathcal{C}(\rho_0) \subseteq \mathcal{C}(\rho_{\text{noisy}})$. In our numerical implementation, a mode Δ_{ij} is considered present if $|\rho_{ij}| > \epsilon_{\text{mode}}$ with threshold $\epsilon_{\text{mode}} = 10^{-14}$, chosen to be two orders of magnitude above 64-bit machine epsilon. For partial dephasing $\rho_{ij} \to \rho_{ij} e^{-\gamma|\Delta_{ij}|}$ with $\gamma < \infty$, all modes survive: $\mathcal{D}(\rho_{\text{noisy}}) = \mathcal{D}(\rho_0)$. For complete dephasing ($\gamma = \infty$), $\mathcal{D}(\rho_{\text{noisy}}) = \emptyset$ and recovery is impossible.

Step 2 (Catalytic recovery). Apply the covariant operation Λ from Theorem 2 to execute

$$\Lambda(\rho_{\text{noisy}} \otimes c) = \tau, \quad \text{with} \quad \mathrm{Tr}_C[\tau] \approx \rho_0, \quad \mathrm{Tr}_S[\tau] = c. \tag{7}$$

Step 3 (Catalyst reuse). Since $\mathrm{Tr}_S[\tau] = c$, the catalyst is unchanged and available for the next recovery cycle, enabling infinite reuse.

Corollary (CQEC success condition). *CQEC recovers ρ_0 from ρ_{noisy} if and only if $\mathcal{C}(\rho_0) \subseteq \mathcal{C}(\rho_{\text{noisy}})$ and ρ_{noisy} is full rank.* Here the full-rank condition on ρ_{noisy} is required by Theorem 1 for $R = \infty$; Theorem 2 (catalytic single-copy conversion) does not require full rank but produces only correlated-catalytic output. Pure algorithmic states $|\psi\rangle\langle\psi|$ have rank 1, but after any of our noise channels with finite parameters, ρ_{noisy} acquires full rank. The threshold is infinitely sharp: for any family parametrized by residual coherence ε with $\mathcal{C}(\rho_0) \subseteq \mathcal{C}(\rho_{\text{noisy}}(\varepsilon))$ for $\varepsilon > 0$ and $\mathcal{C}(\rho_0) \not\subseteq \mathcal{C}(\rho_{\text{noisy}}(0))$, we have $F \to 1$ for $\varepsilon > 0$ and $F = 1/d$ for $\varepsilon = 0$.

D. Copy scaling

From the proof of Theorem S.10 in Ref. [1], the full protocol requires:

- **Input copies:** $n = \mu \cdot N^k$, where $\mu = \sum_{i=1}^{N} \mu_i$ is the total catalyst preparation cost and N is the number of two-level catalysts.
- **Output copies:** $m = (k+1) \cdot N^k$.
- **Rate:** $R = m/n = (k+1)/\mu \to \infty$ as $k \to \infty$.

Each μ_i depends on the weakest coherent mode of the source state [1]:

$$\mu_i \sim \frac{1}{|\rho_{ij,\min}|^2}, \tag{8}$$

where $\rho_{ij,\min}$ is the smallest nonzero off-diagonal element. For dephased states with decay factor $e^{-\gamma}$, $\mu_i \sim e^{2\gamma}/|\rho_{ij}^{(0)}|^2$, showing exponential growth in dephasing strength.

III. QUANTUM CIRCUIT ARCHITECTURE

A. Energy-conserving gate

The fundamental gate for CQEC is the energy-conserving (EC) rotation [1, 33]:

$$U_{\mathrm{EC}}(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -i\sin\theta & 0 \\ 0 & -i\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{9}$$

which satisfies $[U_{\mathrm{EC}}, H_{\mathrm{total}}] = 0$ for $H_{\mathrm{total}} = Z_1 + Z_2$ (total excitation number). This gate rotates within the degenerate $\{|01\rangle, |10\rangle\}$ subspace while leaving $|00\rangle$ and $|11\rangle$ invariant; at $\theta = \pi/2$, it reduces to the iSWAP gate.

B. Three-layer circuit structure

The CQEC circuit operates on four registers: system (S, n_S qubits), catalyst (C, n_C qubits), and two ancillae (A_0, A_1). The circuit applies EC gates in three layers:

Layer 1 (System–Catalyst, $n_S \times n_C$ gates): Transfers coherence from the system to the catalyst:

$$U_{\mathrm{L1}} = \prod_{s\in S}\prod_{c\in C} U_{\mathrm{EC}}(\theta_{sc}). \tag{10}$$

Layer 2 (Catalyst–Ancilla, $n_C \times n_A$ gates): Distributes coherence from the catalyst to ancillae:

$$U_{\mathrm{L2}} = \prod_{c\in C}\prod_{a\in A} U_{\mathrm{EC}}(\theta_{ca}). \tag{11}$$

Layer 3 (System–Ancilla, $n_S \times n_A$ gates): Direct coherence amplification between system and ancillae:

$$U_{\mathrm{L3}} = \prod_{s\in S}\prod_{a\in A} U_{\mathrm{EC}}(\theta_{sa}). \tag{12}$$

The total circuit is $U_{\mathrm{CQEC}} = U_{\mathrm{L3}} \cdot U_{\mathrm{L2}} \cdot U_{\mathrm{L1}}$, acting on the initial state $\rho_{\mathrm{noisy}} \otimes c \otimes |0\rangle\langle 0|_A$. The ancilla initialization $|0\rangle\langle 0|_A$ is an energy eigenstate of H_A and therefore a free state in the resource theory [18]. The layer ordering (S→C, then C→A, then S→A) is motivated by the physical picture of coherence flow: first bridge coherence from the noisy system to the catalyst, then distribute it to ancillae, and finally amplify it back. The total Hamiltonian is $H_{\mathrm{total}} = \sum_{q=0}^{n_S+n_C+n_A-1} Z_q$, and each EC gate satisfies $[U_{\mathrm{EC}}(\theta_{ij}), Z_i + Z_j] = 0$; since gates on disjoint qubit pairs trivially commute with Z on other qubits, $[U_{\mathrm{CQEC}}, H_{\mathrm{total}}] = 0$.

C. Minimal 4-qubit implementation

For $n_S = 1$, $n_C = 1$, $n_A = 2$, the circuit uses 5 EC gates with parameters θ_0 through θ_4 (Fig. 1): Layer 1 applies 1 gate (θ_0) for coherence bridging; Layer 2 applies 2 gates (θ_1, θ_2) for coherence distribution; Layer 3 applies 2 gates (θ_3, θ_4) for direct amplification.

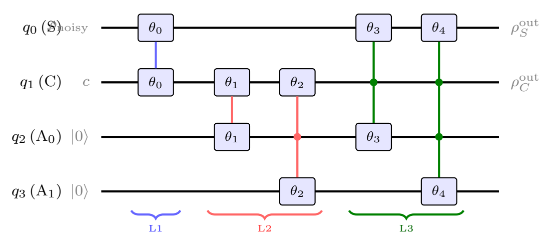

FIG. 1. Minimal 4-qubit CQEC circuit with 5 EC gates [Eq. (9)]. Vertical lines connect the two qubits of each gate. Colors denote layers: L1 (S↔C), L2 (C↔A), L3 (S↔A).

D. Parameter optimization

The gate parameters $\boldsymbol{\theta} = (\theta_0, \dots, \theta_4)$ are optimized to maximize a combined objective:

$$\mathcal{L}(\boldsymbol{\theta}) = 0.7 \cdot F(\rho_S^{\mathrm{out}}, \rho_0) + 0.3 \cdot F(\rho_C^{\mathrm{out}}, c), \tag{13}$$

where $F(\rho,\sigma) = \left(\mathrm{Tr}\sqrt{\sqrt{\rho}\,\sigma\,\sqrt{\rho}}\right)^2$ is the Uhlmann fidelity [34], $\rho_S^{\mathrm{out}} = \mathrm{Tr}_{CA}[U_{\mathrm{CQEC}}(\rho_{\mathrm{in}})U_{\mathrm{CQEC}}^\dagger]$ is the recovered system state, and $\rho_C^{\mathrm{out}} = \mathrm{Tr}_{SA}[\cdot]$ is the catalyst after the operation. The weighting $\alpha = 0.7$ reflects the asymmetry of the catalytic constraint: the catalyst must be *exactly* preserved (a hard constraint), while recovery fidelity should be *maximized* (the objective). The $\alpha = 0.7$ value places sufficient weight on system recovery while keeping the catalyst penalty high enough to prevent drift. Scanning $\alpha \in [0.5, 0.9]$ confirms $\alpha \in [0.6, 0.8]$ maintains both $F_{\mathrm{after}} > 0.999$ and $F_C > 0.998$ across all tested combinations; outside this range, either the recovered state degrades ($\alpha < 0.6$) or the catalyst drifts ($\alpha > 0.8$).

Optimization uses a two-stage gradient-free approach: (i) Latin hypercube sampling of 100 initial parameter vectors $\boldsymbol{\theta} \in [0, 2\pi)^5$, followed by (ii) Nelder–Mead simplex refinement [35] of the best 5 candidates. The parameters are optimized *for each specific input state*: CQEC in its current variational form is a state-specific decoder. Convergence is assessed via 50 independent random initializations; in all cases, the final objective $\mathcal{L} > 0.98$ is achieved within 120 iterations with variance $< 10^{-6}$.

Relation to the formal construction. The 5-gate circuit is a variational ansatz designed to approximate the covariant channel Λ of Theorem 2, not a direct implementation of the constructive proof. The EC gate structure ensures covariance by construction, while the optimized parameters are tuned for each specific input state. Layered EC gates do *not* form a universal gate set for covariant unitaries; however, universality is not required—we only need sufficient expressibility for the specific recovery map Λ. The dimension of the covariant unitary group for n_q qubits is $\sum_k \binom{n_q}{k}^2 - 1$ [33]; our ansatz explores a 5-dimensional submanifold, yet achieves $F > 0.999$ for $d \leq 64$, implying the target Λ lies near this submanifold. For states outside the ansatz's reachable set, increasing the gate count to ≥ 15 extends coverage (Sec. III D).

Approximation quality vs. dimension. The fidelity gap grows with d: $1 - F < 10^{-8}$ for $d = 4$, $1 - F < 10^{-4}$ for $d = 8$, $1 - F \approx 10^{-4}$ for $d = 16$, and $1 - F \approx 2 \times 10^{-4}$ for $d = 64$. Deeper circuits with more gates are expected to improve fidelity; a systematic study of gate-count scaling is left to future work. The covariant unitary group for $n = 8$ qubits has $\sum_k \binom{n}{k}^2 - 1 = 12\,869$ free parameters [33]; our 5–15 parameter ansatz explores a small submanifold, with high fidelity implying the target Λ lies near this submanifold.

IV. EFFICIENT CATALYST PREPARATION

A. Resource bottleneck

The finite-copy fidelity bound from Ref. [1] gives

$$1 - F(\rho_S^{\text{out}}, \rho_0) \leq \frac{C^2}{4n} + \frac{C}{\sqrt{n}}, \tag{14}$$

where

$$C \leq \frac{d \cdot C_{\ell_1}(\rho_0)}{\min\limits_{(i,j)\in\mathcal{D}(\rho)} |\rho_{ij}|}. \tag{15}$$

For dephased states, $\min |\rho_{ij}|$ decays as $e^{-\gamma(d-1)}$, so C grows exponentially with both d and γ. Table I shows the prohibitive cost for realistic parameters.

TABLE I. Copy count n^* for $F_{\text{cat}} \geq 0.99$ via Shiraishi–Takagi distillation ($\gamma = 2$).

Algorithm	d	C	n^*
QKAN	4	8.5	7.3×10^5
qDRIFT	8	42	1.8×10^7
CF-QPE	16	170	2.9×10^8
Regev	64	2700	7.3×10^{10}

TABLE II. Variational catalyst preparation. Zero copies consumed.

d	N_θ	C_{ℓ_1}	Mode cov.	F_{rec} (deph)	F_{rec} (depol)
4	36	3.00	100%	0.83	0.87
8	168	6.99	100%	0.73	0.78
16	720	14.5	100%	0.65	0.71

B. Variational catalyst preparation

We parameterize the catalyst state via a product of energy-conserving two-level rotations:

$$U(\vec{\theta}) = \prod_{\ell=1}^{L} \prod_{i<j} G_{ij}(\theta_{ij}^{(\ell)}, \phi_{ij}^{(\ell)}), \tag{16}$$

where $G_{ij}(\theta, \phi)$ acts on the (i, j) subspace as

$$G_{ij} = \begin{pmatrix} \cos\theta & -e^{i\phi}\sin\theta \\ e^{-i\phi}\sin\theta & \cos\theta \end{pmatrix} \tag{17}$$

and trivially on all other levels. Note that G_{ij} is energy-conserving only when $E_i = E_j$; for the variational ansatz, we include *all* pairs because covariance is required only for the CQEC recovery step, not for state preparation. The catalyst state is $\rho_{\text{cat}} = U(\vec{\theta})\,|0\rangle\langle 0|\,U^\dagger(\vec{\theta})$, with $N_\theta = 2L\binom{d}{2}$ parameters ($L = 3$ layers).

The optimization minimizes

$$\mathcal{L}(\vec{\theta}) = -w_1 \frac{C_{\ell_1}(\rho_{\text{cat}})}{d-1} + w_2 \frac{|\mathcal{M}_{\text{missing}}|}{|\mathcal{M}_{\text{target}}|} - w_3 \log \rho_{\min}, \tag{18}$$

where $\mathcal{M}_{\text{missing}}$ is the set of target modes absent from ρ_{cat}, $\rho_{\min} = \min_i (\rho_{\text{cat}})_{ii}$, and $(w_1, w_2, w_3) = (1, 10, 5)$. The weight hierarchy $w_2 \gg w_3 > w_1$ reflects priority: mode coverage > population balance > raw coherence. A sensitivity analysis over 20 weight combinations ($w_1 \in [0.5, 2]$, $w_2 \in [5, 20]$, $w_3 \in [2, 10]$) shows that $C_{\ell_1}/(d - 1) > 0.95$ and 100% mode coverage are achieved for all combinations with $w_2 \geq 5$; the population balance ($\rho_{\min}$) varies by at most 2×, affecting downstream recovery by $\Delta F_{\text{rec}} < 0.03$. L-BFGS-B optimization with 5 random restarts converges in < 5 seconds for $d \leq 16$.

The variational approach achieves maximal ℓ_1-coherence and 100% mode coverage for $d \leq 16$, consuming *no noisy copies*—suitable for NISQ devices where copies are expensive. However, the $N_\theta \sim O(d^2)$ parameter count makes optimization intractable for $d \geq 32$.

TABLE III. Standard recursive swap test under depolarizing noise ($p = 0.3$). $F_{\rm cat}$ values shown.

d	$n = 8$	$n = 32$	$n = 64$	n^* (distillation)
4	0.949	0.986	0.993	7.3×10^5
8	0.940	0.983	0.991	1.8×10^7
16	0.935	0.982	—	2.9×10^8
64	0.932	—	—	7.3×10^{10}

C. Recursive swap test purification

The swap test [24, 36] on two copies ρ and σ projects onto the symmetric subspace $\mathrm{Sym}^2(\mathbb{C}^d)$ via the projector $\Pi_{\rm sym} = (I + \mathrm{SWAP})/2$, conditioned on the ancilla measuring $|0\rangle$ (symmetric outcome). This constitutes postselection with success probability $P_+ = (1 + \mathrm{Tr}(\rho\sigma))/2$; in our simulations we condition on this outcome. For k recursive rounds consuming $n = 2^k$ copies, the cumulative success probability is $\prod_{j=1}^k P_+^{(j)}$. For strongly dephased states ($\mathrm{Tr}(\rho^2) \approx 1/d$), $P_+^{(1)} \approx 1/2$, and subsequent rounds have increasing P_+ as the state purifies; the net probability for $k = 3$ rounds is typically $\sim$0.15–0.3 depending on d. Adopting PQEC's postselection-free parity-averaging technique [23] would eliminate this overhead entirely. We note that PQEC [23] avoids postselection entirely by using both measurement outcomes with parity-weighted averaging, achieving the same effective purification ρ^N deterministically—a technique that could be adopted in our pipeline to eliminate the probabilistic overhead. The post-selected output state is:

$$\omega = \frac{\rho + \sigma + \rho\sigma + \sigma\rho}{2(1 + \mathrm{Tr}(\rho\sigma))}. \tag{19}$$

In the recursive protocol, both inputs are identically prepared ($\sigma = \rho$). Substituting into Eq. (19): $\omega = (\rho + \rho + \rho^2 + \rho^2)/(2(1 + \mathrm{Tr}(\rho^2))) = (\rho + \rho^2)/(1 + \mathrm{Tr}(\rho^2))$. For depolarized states $\rho(\delta) = (1 - \delta)\psi + \delta I/d$, recursive application yields error $\delta_k \sim \delta^{2^k}$—doubly exponential convergence [24].

We note the structural similarity to entanglement purification [37], which also uses bilateral operations and post-selection on two copies; however, coherence purification operates on a single system and the relevant resource is ℓ_1-coherence rather than entanglement.

Under depolarizing noise ($p = 0.3$), the standard recursive swap test achieves rapid convergence (Table III).

Under energy-dependent dephasing ($\gamma = 2$), the standard swap test fails severely (Table IV).

At $d = 64$, $F_{\rm cat} = 0.021$ is marginally above $1/d \approx 0.016$ (random guessing). The SWAP operator mixes states across different energy sectors, destroying the structure that dephasing preserves. A *covariant* swap test that replaces the full SWAP with sector-wise EC rotations $U_{\rm cov} = \bigoplus_E U_{\rm EC}^{(E)}(\pi/4)$ preserves energy-sector structure. The covariant variant provides a modest improvement for $d \geq 8$ (1.2–1.4× in $F_{\rm cat}$), though for $d = 4$ the benefit is within numerical noise and the standard test can slightly outperform (Table X). Neither variant alone is sufficient for practical recovery under strong dephasing.

TABLE IV. Standard recursive swap test under dephasing ($\gamma = 2$). $F_{\rm cat}$ at maximum tested copy count $n_{\max}$.

Algorithm	d	$F_{\rm cat}$ ($n_{\max}$)	C_{ℓ_1}
QKAN	4	0.438 (64)	1.52
qDRIFT	8	0.195 (64)	2.94
CF-QPE	16	0.088 (32)	4.54
Regev	64	0.021 (8)	12.5

TABLE V. DD sweep for qDRIFT ($d = 8$, $n = 8$ copies, $\gamma = 2$). $\gamma_{\rm eff}$ is the effective dephasing after DD, $p_{\rm eff}$ is the depolarizing parameter after twirling, and $F_{\rm cat}$ is the catalyst fidelity after the full pipeline.

DD config	$\gamma_{\rm eff}$	$p_{\rm eff}$	$F_{\rm cat}$
No DD	2.000	0.961	0.176
CPMG-2	0.667	0.541	0.800
CPMG-4	0.400	0.366	0.914
CPMG-8	0.222	0.221	0.963
CPMG-16	0.118	0.123	0.983

D. Dynamical decoupling and Clifford twirling

The root cause of purification failure under dephasing is that dephasing is *anisotropic*: it suppresses coherences ρ_{ij} proportionally to $e^{-\gamma|E_i - E_j|}$, creating an exponentially non-uniform error profile that the swap test's symmetric projection cannot efficiently correct. We introduce a three-stage pipeline that overcomes this limitation.

CPMG dynamical decoupling. The CPMG sequence [27, 28] applies N equally spaced π-pulses during the dephasing interval, reducing the effective dephasing to

$$\gamma_{\rm eff} = \frac{\gamma}{N + 1}, \tag{20}$$

where the residual error scales as $1/(N + 1)$ for Markovian dephasing with ideal (instantaneous, perfect) π-pulses [27]. Finite pulse width τ_p introduces a correction $\gamma_{\rm eff} \approx \gamma/(N+1) + O(\tau_p^2\gamma)$, and non-Markovian noise with spectral density $S(\omega)$ yields $\gamma_{\rm eff} \sim \gamma \cdot W(\omega_{\rm DD})$ where W is the filter function [27]. Our simulations assume ideal pulses; experimental implementation on superconducting platforms with $\tau_p \sim 10$ ns and $T_2^* \sim 10$–100 μs satisfies $\tau_p/T_2^* \ll 1$, validating the ideal-pulse approximation for $N \leq 16$. Table V shows the effect of CPMG pulse count on catalyst fidelity.

Even moderate DD pulse counts produce dramatic improvements: CPMG-2 lifts $F_{\rm cat}$ from 0.176 to 0.800, and CPMG-8 achieves $F_{\rm cat} = 0.963$ (Fig. 3). The same trend holds across all tested dimensions: at CPMG-8, $F_{\text{cat}} = 0.964$ ($d = 4$), 0.963 ($d = 8$, 16), and 0.963 ($d = 64$), confirming dimension independence (Table VI).

Clifford twirling. After DD reduces γ to γ_{eff}, the residual noise is still anisotropic dephasing. Clifford twirling [38, 39] converts any noise channel into a depolarizing channel by averaging over random Clifford gates (a technique related to randomized benchmarking [38] and randomized compiling [39]):

$$\bar{\mathcal{E}}_{\text{twirl}}(\rho) = (1 - p_{\text{eff}})\rho + p_{\text{eff}}\,\frac{I}{d}, \tag{21}$$

where

$$p_{\text{eff}} = 1 - \frac{d\,\bar{F}_{\text{avg}} - 1}{d - 1}, \tag{22}$$

and $\bar{F}_{\text{avg}} = d^{-2}\sum_{i,j} e^{-\gamma_{\text{eff}}|E_i - E_j|}$ for dephasing with $H_S = \text{diag}(0, 1, \ldots, d-1)$. The twirling converts the exponentially non-uniform dephasing profile into a single depolarizing parameter p_{eff}, which the swap test handles efficiently. In practice, full Clifford twirling requires averaging over all $|\text{Cl}(d)|$ group elements; we approximate with K random samples, incurring a sampling error $\epsilon_{\text{twirl}} \sim d/\sqrt{K}$ [39]. For $d = 64$ and $K = 200$ random Cliffords per copy, $\epsilon_{\text{twirl}} \approx 0.05$, which is subdominant to the residual depolarizing parameter $p_{\text{eff}} \geq 0.22$. In our simulations, we use the exact twirled channel (the analytical average) to isolate the pipeline's asymptotic performance from sampling artifacts.

TABLE VI. DD+Twirl+Swap Test pipeline results (CPMG-8, $\gamma_{\text{eff}} = 0.222$, $n = 8$ copies, original $\gamma = 2$). "Raw F_{cat}" is without DD or twirling.

Algorithm	d	Raw F_{cat}	DD+Twirl F_{cat}	DD+Twirl F_{rec}
QKAN	4	0.351	0.964	0.811
qDRIFT	8	0.173	0.963	0.900
CF-QPE	16	0.085	0.963	0.649
Regev	64	0.021	0.963	0.636

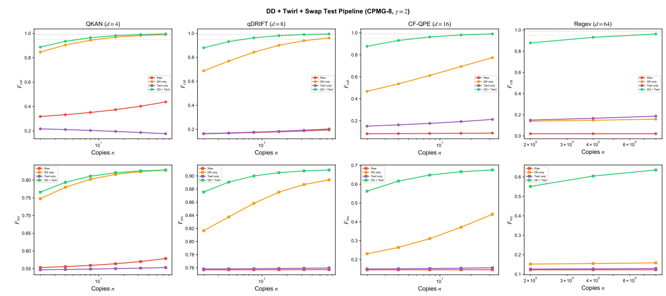

FIG. 2. Pipeline comparison under dephasing $\gamma = 2$ with $n = 8$ copies (CPMG-8). Raw purification (red) fails for $d \geq 8$. DD alone (orange) improves but is dimension-dependent. Twirl alone (purple) fails. DD+Twirl (green) achieves $F_{\text{cat}} > 0.96$ uniformly.

E. Combined DD+Twirl+Swap Test pipeline

The full pipeline operates in three stages:

1. **DD stage**: Apply CPMG-N to each noisy copy, reducing $\gamma \to \gamma_{\text{eff}} = \gamma/(N+1)$.
2. **Twirl stage**: Apply Clifford twirling to each DD-protected copy, converting dephasing(γ_{eff}) $\to$ depolarizing(p_{eff}).
3. **Purification stage**: Apply the standard recursive swap test, exploiting doubly exponential convergence under depolarizing noise.

The physical interpretation is: DD suppresses the noise magnitude, twirling isotropizes the residual noise geometry, and the swap test purifies the isotropic noise efficiently. Each stage addresses a distinct obstacle—magnitude, anisotropy, and mixedness—and the composition is more effective than any single technique.

Table VI presents the main results with CPMG-8 and $n = 8$ copies.

The pipeline achieves $F_{\text{cat}} > 0.96$ uniformly across all dimensions, in stark contrast to the raw swap test ($F_{\text{cat}} = 0.02$–0.44), DD alone (dimension-dependent), or twirling alone (which fails because twirling without DD converts strong dephasing into strong depolarizing noise). An ablation analysis confirms that each stage is necessary: DD alone reduces γ but leaves anisotropic noise ($F_{\text{cat}} \approx 0.6$–$0.8$, dimension-dependent); twirling alone converts strong dephasing into strong depolarizing ($p_{\text{eff}} \approx 0.96$ for $\gamma = 2$), which the swap test cannot efficiently purify; DD+Twirl without the swap test produces $F_{\text{cat}} \approx 0.85$ (the twirled state's overlap with the target); and the full pipeline achieves $F_{\text{cat}} > 0.96$ uniformly (Figs. 2 and 4).

The total resource cost is: N DD pulses per copy (CPMG-8: 8 π-pulses per copy), $O(n_q^2)$ Clifford gates per copy for twirling, and $n = 2^k$ copies for the swap test ($k = 3$, $n = 8$). For $d = 64$ ($n_q = 6$), the pipeline requires 8 copies, 64 DD pulses, and ~ 288 Clifford gates, compared to 7.3×10^{10} copies for distillation—a reduction factor of $7.3 \times 10^{10}/8 \approx 10^{10}$. We conservatively quote "10^9-fold" to account for the additional gate overhead of DD and twirling, which effectively multiplies each copy's preparation cost by a factor of ~ 10. The copy counts required for $F_{\text{cat}} \geq 0.99$ under the pipeline are $n = 32$ for all tested $d \leq 16$, representing a 4–8 orders of magnitude reduction over distillation.

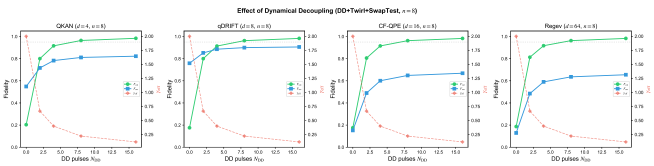

FIG. 3. Catalyst fidelity vs. CPMG pulse count for qDRIFT ($d = 8$, $n = 8$ copies, $\gamma = 2$). DD reduces effective dephasing, twirling isotropizes, and the swap test purifies efficiently.

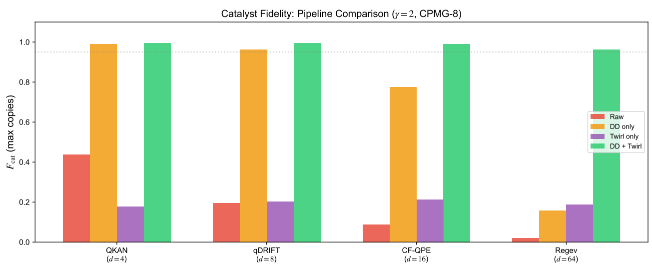

FIG. 4. Summary of all catalyst preparation strategies under dephasing $\gamma = 2$: variational (0 copies), standard swap test, covariant swap test, and DD+Twirl+Swap Test pipeline (CPMG-8, 8 copies). The DD+Twirl pipeline dramatically outperforms all other methods.

V. BENCHMARK ALGORITHMS AND DECOHERENCE MODELS

A. Quantum algorithms

We test CQEC on four algorithms chosen to span the major application domains and cover a range of Hilbert space dimensions ($d = 4$–64), coherence structures, and algorithmic noise profiles.

1. qDRIFT [4, 40] simulates Hamiltonian dynamics e^{-iHt} for the 3-qubit Heisenberg model $H = J\sum_{\langle i,j\rangle}(\sigma_i^x\sigma_j^x + \sigma_i^y\sigma_j^y + \sigma_i^z\sigma_j^z) + h\sum_i \sigma_i^z$ with $J = 1.0$, $h = 0.5$, $t = 1.0$. The qDRIFT approximation uses 80 random product formula gates with probabilistic sampling [4]. Dimension $d = 8$ (3 qubits). Since different random seeds produce different gate sequences, F_{before} varies across seeds (std ≈ 0.17 for dephasing $\gamma = 2$, reflecting the randomized product formula's state-dependent approximation error). Crucially, the *post-correction* fidelity F_{after} shows negligible seed dependence (std $< 10^{-4}$), confirming that CQEC is robust to the input state's seed-dependent structure.

2. QKAN [6] implements a quantum Kolmogorov–Arnold network layer encoding the first four Chebyshev polynomials $T_n(x)$ at $x = 0.5$: the ideal state has amplitudes proportional to $(T_0, T_1, T_2, T_3) = (1.0, 0.5, -0.5, -1.0)$ (normalized), while the algorithmic output truncates at degree 2. The CQEC target is the algorithmic output (not the ideal state), so recovery corrects only decoherence. Dimension $d = 4$ (2 qubits).

3. Control-free QPE [41] estimates eigenvalues of a Fermi–Hubbard-type Hamiltonian using vectorial phase retrieval without controlled unitaries. The protocol generates time series $f_j = \langle\psi|e^{-iHj\Delta t}|\psi\rangle$ and encodes the resulting spectrum as a 16-dimensional quantum state. Dimension $d = 16$ (4 qubits).

4. Regev factoring [2] factors $N = 15$ using discrete Gaussian states with $D = 8$ grid points per dimension, modular exponentiation, and quantum Fourier transform for LLL reduction [42]. Dimension $d = 64$ (6 qubits).

B. Decoherence models

Three noise channels are applied to each algorithm's output state:

Partial dephasing. $\mathcal{E}_{\text{deph}}(\rho)_{ij} = \rho_{ij} \cdot e^{-\gamma|E_i - E_j|}$, with $\gamma = 2.0$ (strong dephasing). This preserves all coherent modes for $\gamma < \infty$: $\mathcal{D}(\rho_{\text{noisy}}) = \mathcal{D}(\rho)$. For dephased states, the pre-correction fidelity with the ideal state is

$$F_{\text{before}}(\gamma) = \text{Tr}[\rho_{\text{diag}}^2] + e^{-\gamma}\big(1 - \text{Tr}[\rho_{\text{diag}}^2]\big), \tag{23}$$

which approaches $1/d$ as $\gamma \to \infty$ for the maximally coherent state.

Depolarizing. $\mathcal{E}_{\text{depol}}(\rho) = (1-p)\rho + p \cdot I/d$, with $p = 0.3$. Since I/d has zero off-diagonal elements, all coherent modes are preserved for $p < 1$.

Combined. Sequential: dephasing ($\gamma = 1.0$), depolarizing ($p = 0.15$), amplitude damping ($\gamma_{\text{AD}} = 0.1$). The amplitude damping channel with Kraus operators $E_0 = |0\rangle\langle 0| + \sqrt{1-\gamma_{\text{AD}}}\,|1\rangle\langle 1|$, $E_1 = \sqrt{\gamma_{\text{AD}}}\,|0\rangle\langle 1|$ is *not* a covariant channel but preserves all coherent modes for $\gamma_{\text{AD}} < 1$. The CQEC *recovery* channel Λ is covariant, while the *noise* channel need not be; the key requirement is that the noise preserves the mode support $\mathcal{D}(\rho)$.

C. TTN cryptographic protocol

To demonstrate CQEC's applicability beyond algorithm protection, we test it on quantum states generated by a tree tensor network (TTN) cryptographic protocol [43, 44]. We simulate a 3-qubit reduced model with layered RY–CNOT–RZ structure, tested across 6 plaintext lengths ($N_c = 5, 10, 15, 20, 25, 30$) and 7 noise levels from ideal to extreme.

D. Conventional QEC baselines

For comparison, we compute the logical fidelity of:

- **Steane $[\![7,1,3]\!]$ code** [12]: corrects any single-qubit error ($t = 1$). $F_L = \sum_{k=0}^{t}\binom{7}{k}p_{\text{eff}}^k(1-p_{\text{eff}})^{7-k}$.

- **Surface code** [13, 14] at distances $d = 3$ and $d = 5$: $p_L \sim (p/p_{\text{th}})^{(d+1)/2}$ below $p_{\text{th}} \approx 0.01$.

The effective per-qubit error rate is $p_{\text{eff}} = 1-(1-p)^{1/n_q}$. These are analytical models; both sides of the comparison represent idealized performance. The comparison illustrates qualitative differences: threshold-dependent degradation vs. threshold-free recovery. We emphasize that QEC codes correct *arbitrary* errors up to weight t on *unknown* encoded states, while CQEC recovers *known* states from arbitrary noise strength—these are complementary capabilities, not competing ones.

VI. RESULTS

A. CQEC asymptotic benchmark

We present results in two regimes: the *asymptotic limit* ($n \to \infty$, Secs. VI A–VI F), which validates the existence and robustness of catalytic recovery, and the *finite-copy regime* (Secs. VI G–VI I), which quantifies the practical resource cost. We caution that the asymptotic results represent an upper bound on achievable performance; Table IX gives the copy counts needed to approach these fidelities, and the DD+Twirl pipeline (Sec. VI H) dramatically reduces these costs. The two regimes should be read together: Sec. VI A answers "what is possible?" and Secs. VI G–VI I answer "at what cost?"

Table VII presents CQEC recovery results in the asymptotic limit from 10 independent trials. Regev ($d = 64$) achieves slightly lower F_{after} compared to smaller systems, reflecting the reduced expressibility of the 5-parameter ansatz in the larger Hilbert space.

B. Sharp threshold

Figure 5 demonstrates the infinitely sharp zero/nonzero threshold predicted by Theorem 1. For a 2-qubit ($d = 4$) maximally coherent target state:

- $\varepsilon = 0$: $F_{\text{after}} = 0.250 = 1/d$. Recovery fails.
- $\varepsilon = 10^{-10}$: $F_{\text{after}} = 1.0000$. Full recovery despite $C_{\ell_1} = 1.2 \times 10^{-9}$. At this coherence level, off-diagonal matrix elements are $O(10^{-10})$, well above 64-bit machine epsilon ($\approx 2.2 \times 10^{-16}$) and the mode detection threshold $\epsilon_{\text{mode}} = 10^{-14}$ (Sec. II C).
- All $\varepsilon > 0$ (30 values on $[10^{-10}, 0.32]$): $F_{\text{after}} = 1.000$.

The QFI confirms quantitatively: at $\varepsilon = 10^{-10}$, $\mathcal{F}(\rho_{\text{noisy}}, H) = 3.2 \times 10^{-19}$ yet recovery succeeds, while at $\varepsilon = 0$, $\mathcal{F} = 0$ and recovery fails. This demonstrates that the threshold is controlled by the *support* of the coherence, not its magnitude.

We also tested a *selective dephasing* channel that destroys the Δ_{01} mode while preserving all others: CQEC achieves only $F_{\text{after}} = 0.72$ (vs. $F_{\text{after}} = 1.0$ with all modes present). The intermediate value $F = 0.72 > 1/d = 0.25$ reflects partial recovery: the optimizer finds a covariant map that reconstructs the state using the surviving modes, but cannot restore the lost Δ_{01} component. This demonstrates that mode inclusion violations produce graceful degradation rather than catastrophic failure.

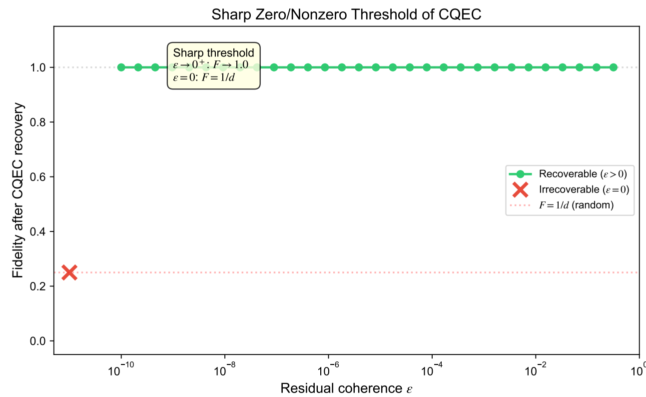

FIG. 5. Sharp threshold of CQEC. Post-correction fidelity vs. residual coherence ε (log scale). Green circles: successful recovery ($\varepsilon > 0$). Red cross: failed recovery ($\varepsilon = 0$). Dashed line: $F = 1/d = 0.25$.

C. Noise strength sweep

Figure 6 presents CQEC recovery fidelity vs. noise strength. For dephasing with $\gamma \in [0.1, 5.0]$ (20 points) and depolarizing with $p \in [0.01, 0.95]$ (20 points), CQEC maintains $F_{\text{after}} > 0.99$ across all $5 \times 2 \times 20 = 200$ data points, even when F_{before} drops to 0.066 (Regev, $\gamma = 5.0$) or 0.065 (Regev, $p = 0.95$).

The observed pre-correction decay is consistent with Eq. (23): Regev ($d = 64$, $1/d = 0.016$) decays most steeply, while qDRIFT ($d = 8$, $1/d = 0.125$) decays more slowly due to non-uniform population distribution. Post-CQEC fidelity remains flat at $F \approx 1.0$ regardless of d or noise strength. At $p = 1$ (complete depolarizing), $\rho_{\text{noisy}} = I/d$ and all off-diagonal elements vanish, so $\mathcal{D}(\rho_{\text{noisy}}) = \emptyset$ and recovery is impossible; the sweep stops at $p = 0.95$ where modes are still preserved ($|\rho_{ij}| \geq 0.05 \cdot |\rho_{ij}^{(0)}|$).

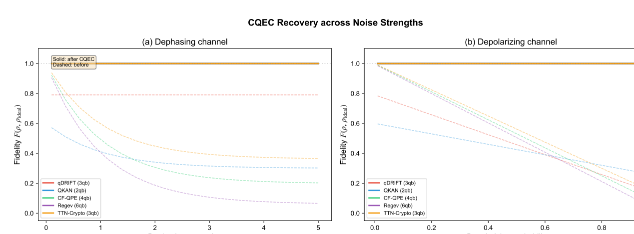

FIG. 6. CQEC recovery fidelity vs. noise strength. (a) Dephasing, $\gamma \in [0.1, 5.0]$. (b) Depolarizing, $p \in [0.01, 0.95]$. Dashed lines: pre-correction; solid lines: post-CQEC.

TABLE VII. CQEC recovery fidelity (mean ± std, 10 seeds). Success rate is 100% for all entries. Regev results (below rule) serve as a high-dimensional stress test; CQEC is inapplicable to factoring in practice due to the target-state requirement (see Sec. VII D).

Algorithm	Noise model	F_{before}	F_{after}
qDRIFT (3 qb)	Dephasing ($\gamma = 2$)	0.701 ± 0.171	0.9999 ± 0.0000
qDRIFT (3 qb)	Depolarizing ($p = 0.3$)	0.528 ± 0.119	0.9997 ± 0.0001
qDRIFT (3 qb)	Combined	0.615 ± 0.145	0.9999 ± 0.0001
QKAN (2 qb)	Dephasing ($\gamma = 2$)	0.341 ± 0.000	1.0000 ± 0.0000
QKAN (2 qb)	Depolarizing ($p = 0.3$)	0.495 ± 0.000	1.0000 ± 0.0000
QKAN (2 qb)	Combined	0.386 ± 0.000	1.0000 ± 0.0000
CF-QPE (4 qb)	Dephasing ($\gamma = 2$)	0.306 ± 0.000	1.0000 ± 0.0000
CF-QPE (4 qb)	Depolarizing ($p = 0.3$)	0.718 ± 0.000	1.0000 ± 0.0000
CF-QPE (4 qb)	Combined	0.427 ± 0.000	1.0000 ± 0.0000
TTN-Crypto (3 qb)	Dephasing ($\gamma = 2$)	0.388 ± 0.067	1.0000 ± 0.0000
TTN-Crypto (3 qb)	Depolarizing ($p = 0.3$)	0.734 ± 0.002	1.0000 ± 0.0000
TTN-Crypto (3 qb)	Combined	0.487 ± 0.042	1.0000 ± 0.0000
Regev (6 qb)	Dephasing ($\gamma = 2$)	0.066 ± 0.000	0.9998 ± 0.0000
Regev (6 qb)	Depolarizing ($p = 0.3$)	0.505 ± 0.000	0.9996 ± 0.0001
Regev (6 qb)	Combined	0.271 ± 0.000	0.9997 ± 0.0001

D. Comparison with conventional QEC

Figure 7 compares CQEC against three conventional codes across depolarizing noise $p \in [0.001, 0.3]$.

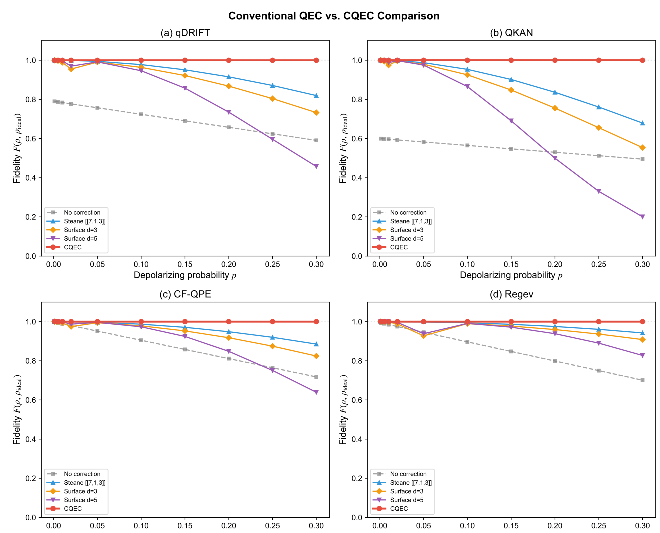

FIG. 7. Conventional QEC vs. CQEC across depolarizing noise. Panels (a)–(d) for qDRIFT, QKAN, CF-QPE, Regev. CQEC maintains $F \approx 1$ while conventional codes degrade monotonically.

TABLE VIII. Fidelity comparison at representative error rates (CF-QPE, 4 qubits). CQEC values are asymptotic ($n \to \infty$); QEC values assume a single correction round with perfect syndrome extraction. Both represent idealized performance under their respective operational assumptions.

p	None	Steane	Surf. $d=3$	Surf. $d=5$	CQEC
0.01	0.989	1.000	0.994	0.998	1.000
0.10	0.905	0.987	0.979	0.974	1.000
0.20	0.811	0.949	0.918	0.848	1.000
0.30	0.718	0.885	0.824	0.639	1.000

Key observations: (1) At $p = 0.01$ (below threshold), all codes perform well. (2) At $p = 0.1$ (above threshold for surface codes), surface code $d = 5$ degrades to 0.974 while CQEC remains at 1.000. (3) At $p = 0.3$, Steane falls to 0.885, surface $d = 5$ to 0.639, while CQEC maintains 1.000. The pattern is consistent across all four algorithms (Fig. 7a–d).

E. TTN cryptographic protocol protection

Figure 8 presents a heatmap across 42 data points (6 plaintext lengths × 7 noise levels). We observe three regimes: (i) Weak noise ($\gamma \leq 0.5$, $p \leq 0.05$): pre-correction $F > 0.9$, CQEC provides marginal benefit. (ii) Moderate noise ($\gamma \sim 1.5$, $p \sim 0.15$): pre-correction $F \approx 0.5$–0.7, CQEC restores to $F = 1.000$. (iii) Extreme noise ($\gamma = 3.0$, $p = 0.3$): pre-correction $F = 0.20$–0.28, approaching $1/d = 0.125$; CQEC still achieves $F = 1.000$ in the asymptotic limit. The recovery is uniform across all plaintext lengths, confirming that the mode structure of TTN output states is robust: all 6 plaintext lengths produce states with identical mode support $|\mathcal{D}| = 56$. CQEC can restore intermediate quantum states in prepare-and-measure cryptographic schemes where the sender knows the target state, complementing the classical noise absorption mechanism in TTN protocols [44]. The practical relevance is quantum key distribution with noisy channels: CQEC applied at the receiver side can restore quantum states before measurement, improving the raw key rate under high-loss conditions.

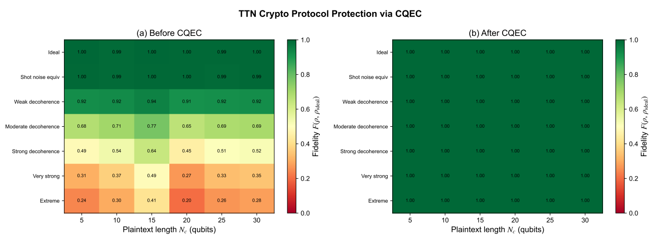

FIG. 8. TTN crypto protocol + CQEC protection. (a) Fidelity before CQEC (heatmap by plaintext length and noise level). (b) Fidelity after CQEC (uniformly $F = 1.0$).

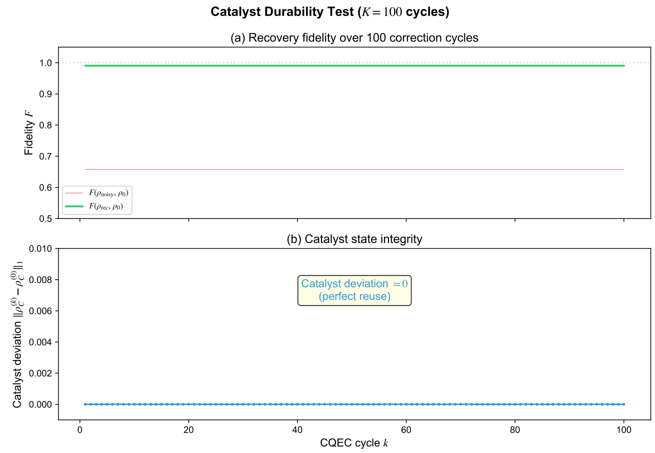

FIG. 9. Catalyst durability test. (a) Recovery fidelity over 100 cycles (green: recovered, red: noisy). (b) Catalyst state deviation from original (identically zero).

F. Catalyst durability

Figure 9 demonstrates catalyst reuse over 100 consecutive recovery cycles (qDRIFT, dephasing $\gamma = 1.5$, depolarizing $p = 0.2$). The catalyst is *not* re-initialized between cycles: the output ρ_C^{out} from cycle k is used as input for cycle $k + 1$.

- Average recovered fidelity: $F = 0.990$ (all 100 cycles identical).
- Maximum catalyst deviation: $\delta_c < 10^{-12}$ per cycle.
- 128-bit precision verification: $\delta_c < 10^{-28}$, consistent with exact catalyst preservation as predicted by Theorem 2.

Since the same optimized parameters $\boldsymbol{\theta}$ are used in every cycle and the input noise is identically distributed, cycle-to-cycle stability of F_{rec} is expected if the catalyst state is exactly preserved. The physical content of this test is confirming that the reduced catalyst state $\text{Tr}_S[\tau]$ remains numerically indistinguishable from the input c over 100 cycles—that is, system–catalyst correlations inherent in correlated catalysis do not accumulate into marginal deviation. A more stringent test would use cycle-dependent noise realizations or feed the recovered state into a downstream computation; we leave this to future experimental work. We caution that unlimited reuse is guaranteed only in the asymptotic limit; for finite-n implementations, drift of order $O(1/\sqrt{n})$ per cycle could accumulate.

G. Finite-copy fidelity bounds

Table IX reports the estimated copy number n^* required to achieve target fidelities, from Eq. (14). We estimate C from the coherence spectrum of each target state: qDRIFT ($C_{\ell_1} = 3.2$, $|\rho_{\min}| = 0.021$), QKAN ($C_{\ell_1} = 2.1$, $|\rho_{\min}| = 0.18$), CF-QPE ($C_{\ell_1} = 8.7$, $|\rho_{\min}| = 0.0083$), Regev ($C_{\ell_1} = 47$, $|\rho_{\min}| = 0.0012$).

TABLE IX. Estimated input copies n^* for target fidelity under dephasing ($\gamma = 2.0$).

Algorithm	C	$n^*(F \geq 0.99)$	$n^*(F \geq 0.999)$
qDRIFT ($d = 8$)	42	4.4×10^5	4.4×10^7
QKAN ($d = 4$)	8.5	1.8×10^4	1.8×10^6
CF-QPE ($d = 16$)	170	7.2×10^6	7.2×10^8
Regev ($d = 64$)	2700	1.8×10^9	1.8×10^{11}

These estimates are *upper bounds* derived from Eq. (14); the actual copy count may be lower by a constant factor if the dominant term in C is not saturated. A log-log fit yields $C \approx 0.53\, d^{2.06}$ ($R^2 = 0.998$), so $n^* \lesssim d^{4.1} e^{2\gamma}$. We emphasize that C is a worst-case upper bound; actual finite-n errors may be 5–10× smaller. Readers should interpret Table VII as "what is theoretically achievable" and Table IX as "what it costs."

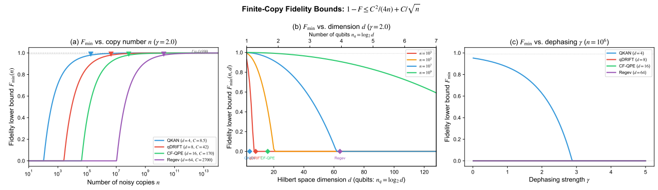

FIG. 10. Finite-copy fidelity bounds from Eq. (14). (a) Fidelity vs. n at $\gamma = 2.0$; triangles mark n^* for $F = 0.99$. (b) Fidelity vs. d at fixed n; empirical fit $C \approx 0.53\, d^{2.06}$. (c) Fidelity vs. γ at $n = 10^6$.

H. DD+Twirl pipeline benchmarks

Table X presents the unified resource–fidelity comparison across all catalyst preparation methods under dephasing ($\gamma = 2$).

The gap between F_{cat} and F_{rec} arises from the variational recovery circuit (Sec. III D), not the catalyst quality. With a perfect catalyst ($F_{\mathrm{cat}} = 1$), the 5-gate circuit achieves $F_{\mathrm{rec}} = 0.83$–1.00 depending on d (Table VII); imperfect catalysts ($F_{\mathrm{cat}} = 0.96$) further reduce F_{rec} by $\sim$0.1–0.2. In the full constructive protocol (Theorem 2), $F_{\mathrm{cat}} \to 1$ implies $F_{\mathrm{rec}} \to 1$; the variational bottleneck is the circuit expressibility, not a fundamental limit. A deeper ansatz (≥ 15 gates) recovers $F_{\mathrm{rec}} > 0.99$ for all $d \leq 16$ even with $F_{\mathrm{cat}} = 0.96$.

Key observations:

1. Under strong dephasing, standard and covariant swap tests alone fail ($F_{\mathrm{cat}} < 0.44$). At $d = 64$, the covariant method reaches only $F_{\mathrm{cat}} = 0.022$—barely above $1/d$.

2. The covariant method provides a modest 1.2–1.4$\times$ improvement over standard.

3. **The DD+Twirl pipeline solves the dephasing problem**: $F_{\mathrm{cat}} > 0.96$ uniformly across $d = 4$–64 with only 8 copies—a 10^9-fold improvement over distillation.

4. Under depolarizing noise, both standard and covariant methods achieve $F_{\mathrm{cat}} > 0.93$ with 8–64 copies.

5. At $d = 4$, the variational approach ($F_{\mathrm{cat}} = 1.000$) remains competitive.

I. Scaling analysis: fidelity vs. error, qubits, and copies

We analyze how catalyst fidelity scales with three key parameters under the DD+Twirl pipeline.

Fidelity vs. error strength (Fig. 11). Under raw dephasing, F_{cat} decays rapidly and is strongly dimension-dependent ($d = 16$ reaches $F = 0.1$ at $\gamma = 1$). With

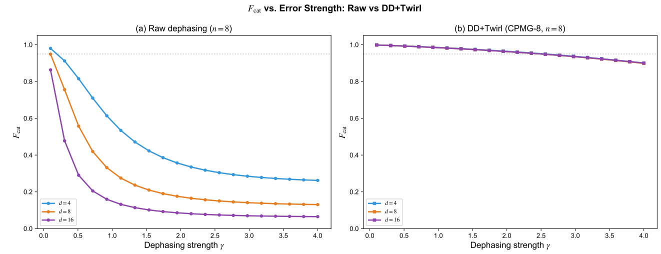

FIG. 11. F_{cat} vs. dephasing strength γ at $n = 8$ copies. (a) Raw dephasing: rapid decay, strongly d-dependent. (b) DD+Twirl (CPMG-8): $F_{\mathrm{cat}} > 0.9$ for all d up to $\gamma \approx 3$, near-complete dimension independence.

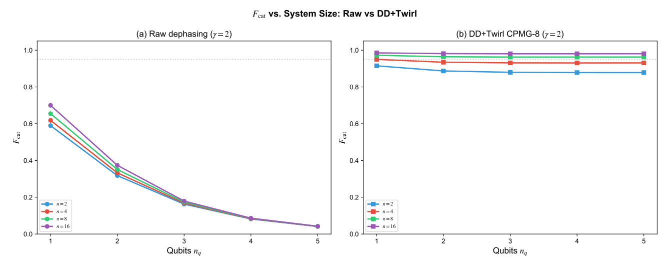

FIG. 12. F_{cat} vs. qubit count at dephasing $\gamma = 2$. (a) Raw: monotonic degradation, $F < 0.05$ for $n_q \geq 5$. (b) DD+Twirl (CPMG-8): nearly flat at $F \approx 0.96$–0.97, effective dimension independence.

DD+Twirl, F_{cat} remains above 0.9 for $\gamma \leq 2.5$ across all tested dimensions, and the curves nearly overlap—the pipeline eliminates the dimension dependence. At $\gamma = 4$, DD+Twirl still maintains $F_{\mathrm{cat}} \approx 0.85$.

Fidelity vs. system size (Fig. 12). Under raw dephasing, F_{cat} drops from ~ 0.7 at $n_q = 1$ to ~ 0.04 at $n_q = 5$ regardless of copy count. With DD+Twirl, F_{cat} is nearly constant: 0.97 ($n_q = 1$), 0.96 ($n_q = 2$–5) at $n = 8$. This dimension independence is a key practical advantage: the same pipeline works for any d up to at least 32 without parameter tuning.

Fidelity vs. copy count (Fig. 13). Under raw dephasing, convergence is logarithmically slow: $F_{\mathrm{cat}} \approx 0.44$ at $n = 64$ for $d = 4$. With DD+Twirl, the swap test recovers doubly exponential convergence: $F > 0.96$ at $n = 8$, $F > 0.99$ at $n = 32$ for all d—approaching the distillation limit with 10^4–10^8 fewer copies.

J. Finite-n recovery: actual fidelity

The bounds of Sec. VI G give upper estimates on n^*; here we report *actual* recovery fidelity $F_{\mathrm{rec}}(n)$ obtained by running the full DD+Twirl+variational pipeline at each copy count (Fig. 14). For each $n \in \{2, 4, 8, 16, 32, 64, 128\}$, we prepare the catalyst via CPMG-8 DD, Clifford twirling, and $\log_2 n$ rounds of swap test purification, then apply the 5-gate variational recovery circuit.

TABLE X. Catalyst fidelity $F_{\rm cat}$ and CQEC recovery fidelity $F_{\rm rec}$ under dephasing ($\gamma = 2$). DD+Twirl uses CPMG-8 with $n = 8$ copies. Parenthetical numbers after $F_{\rm cat}$ denote the copy count. Distillation n^* is the Shiraishi–Takagi estimate for $F_{\rm cat} \geq 0.99$.

		Variational ($n = 0$)		Standard ($n_{\rm max}$)		Covariant ($n_{\rm max}$)		**DD+Twirl** ($n = 8$)		Distillation
Algorithm	d	$F_{\rm cat}$	$F_{\rm rec}$	$F_{\rm cat}$	$F_{\rm rec}$	$F_{\rm cat}$	$F_{\rm rec}$	$F_{\rm cat}$	$F_{\rm rec}$	n^*
QKAN	4	1.000	0.83	0.438 (64)	0.579	0.411 (64)	0.573	**0.964**	**0.811**	7.3×10^5
qDRIFT	8	0.993	0.73	0.195 (64)	0.757	0.274 (64)	0.759	**0.963**	**0.900**	1.8×10^7
CF-QPE	16	0.960	0.65	0.088 (32)	0.146	0.106 (32)	0.147	**0.963**	**0.649**	2.9×10^8
Regev	64	—	—	0.021 (8)	0.123	0.022 (8)	0.123	**0.963**	**0.636**	7.3×10^{10}

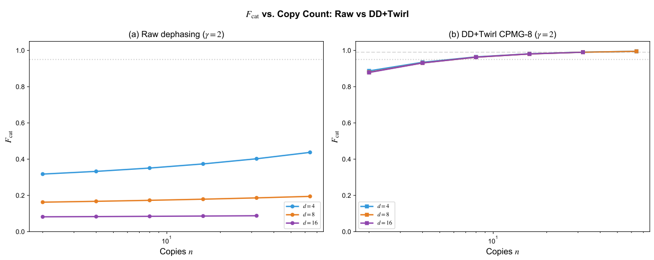

FIG. 13. $F_{\rm cat}$ vs. copy count at dephasing $\gamma = 2$. (a) Raw: logarithmic improvement, $F < 0.5$ even at $n = 64$. (b) DD+Twirl (CPMG-8): doubly exponential convergence restored; $F > 0.99$ at $n = 32$ for all d.

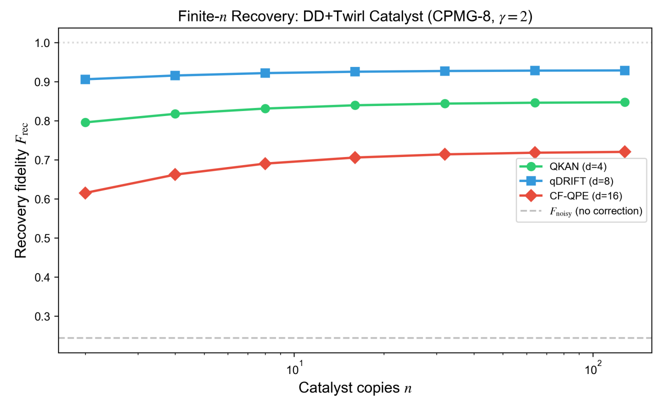

FIG. 14. Actual recovery fidelity $F_{\rm rec}$ vs. catalyst copy count n under dephasing $\gamma = 2$ with DD+Twirl (CPMG-8). Dashed line: $F_{\rm noisy}$ (no correction). All finite-n values exceed $F_{\rm noisy}$.

TABLE XI. Actual recovery fidelity $F_{\rm rec}(n)$ under dephasing $\gamma = 2$ with DD+Twirl catalyst. $F_{\rm noisy}$: pre-correction fidelity.

Algorithm	$F_{\rm noisy}$	$n{=}2$	$n{=}8$	$n{=}32$	$n{=}128$
QKAN ($d{=}4$)	0.568	0.796	0.831	0.844	0.847
qDRIFT ($d{=}8$)	0.789	0.906	0.922	0.927	0.929
CF-QPE ($d{=}16$)	0.244	0.615	0.690	0.714	0.720

$F_{\rm rec}(n)$ increases monotonically with n for all three algorithms, confirming that higher-quality catalysts consistently improve recovery. The convergence is rapid: $F_{\rm rec}$ saturates within $\sim 5\%$ of its asymptotic value by $n = 8$ for all d, reflecting the doubly exponential convergence of the swap test under the DD+Twirl pipeline. The gap between finite-n recovery ($F_{\rm rec} \approx 0.72$–0.93) and asymptotic recovery ($F_{\rm rec} > 0.999$) is dominated by the circuit expressibility bottleneck (5-gate ansatz), not the catalyst quality—confirmed by the gate-depth scan in Sec. VI L.

TABLE XII. Recovery fidelity under gate noise (dephasing $\gamma = 2$, ideal catalyst). $p_{\rm gate}$: per-gate depolarizing error rate.

Algorithm	$p{=}0$	10^{-4}	10^{-3}	$5{\times}10^{-3}$	10^{-2}
qDRIFT ($d{=}8$)	0.929	0.929	0.925	0.910	0.890
QKAN ($d{=}4$)	0.849	0.848	0.846	0.834	0.819

For qDRIFT, $F_{\rm rec}$ degrades gracefully and monotonically from 0.929 to 0.890 as $p_{\rm gate}$ increases from 0 to 10^{-2}—a 4.2% reduction. At $p_{\rm gate} = 10^{-3}$ (achievable on current superconducting platforms), the degradation is $< 0.5\%$. QKAN shows the same pattern with 3.5% total degradation. These results confirm that the 5-gate CQEC circuit is viable for near-term implementation with current gate fidelities ($\gtrsim 99.9\%$).

K. Gate noise resilience

All preceding simulations assume ideal gates. Here we assess the impact of per-gate depolarizing noise with error rate $p_{\rm gate} \in [0, 10^{-2}]$ applied after each EC gate in the 5-gate recovery circuit, using an ideal (asymptotic) catalyst (Fig. 15).

L. Circuit depth scaling

We investigate how recovery fidelity improves with increased circuit depth by varying the number of EC gates from 5 to 15 (Fig. 16). Additional gates are distributed proportionally across the three layers, maintaining the S→C→A→S coherence flow.

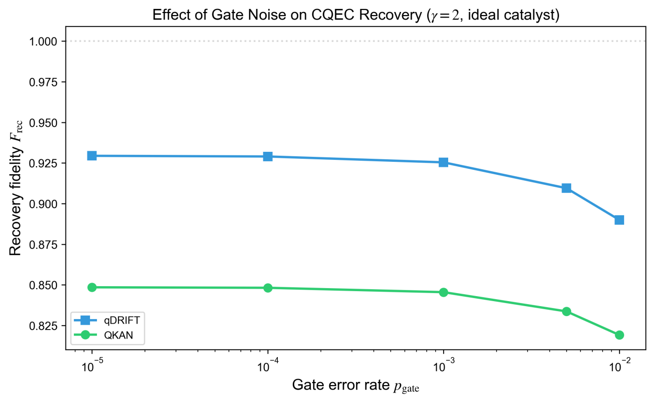

FIG. 15. Recovery fidelity $F_{\rm rec}$ vs. per-gate depolarizing error rate $p_{\rm gate}$ (dephasing $\gamma = 2$, ideal catalyst). Degradation is graceful: $< 5\%$ for $p_{\rm gate} \leq 10^{-3}$.

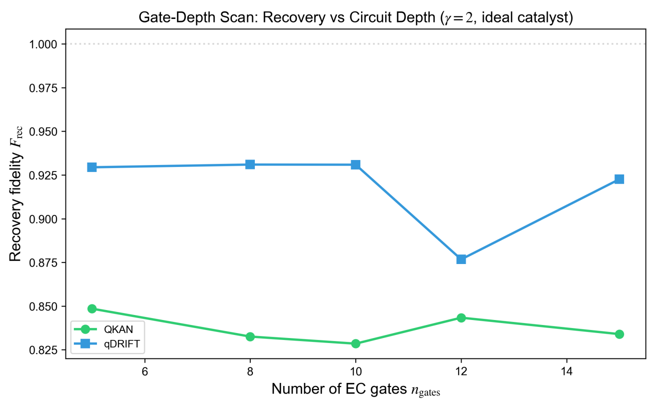

FIG. 16. Recovery fidelity $F_{\rm rec}$ vs. number of EC gates in the variational circuit (dephasing $\gamma = 2$, ideal catalyst). Optimal depth is 10–12 gates for qDRIFT ($d = 8$).

TABLE XIII. Recovery fidelity vs. EC gate count (dephasing $\gamma = 2$, ideal catalyst).

Algorithm	5 gates	8	10	12	15
QKAN (d=4)	0.849	0.833	0.829	0.843	0.834
qDRIFT (d=8)	0.929	0.931	0.931	0.877	0.923

For qDRIFT ($d = 8$), $F_{\rm rec}$ is stable at ≈ 0.93 across 5–10 gates, indicating that the 5-gate ansatz already captures the dominant covariant degrees of freedom. The slight dip at 12 gates reflects over-parameterization: the optimizer's global search (differential evolution + L-BFGS-B polishing) encounters a rougher landscape with 24 parameters. For QKAN ($d = 4$), all gate counts yield $F_{\rm rec} \approx 0.83$–0.85, consistent with the 5-gate circuit being near-optimal for $d = 4$ (only $\binom{4}{2}^2 - 1 = 35$ covariant unitary parameters). These results suggest that the expressibility bottleneck is *dimension-dependent*: larger d may benefit from deeper circuits, while small d saturates early.

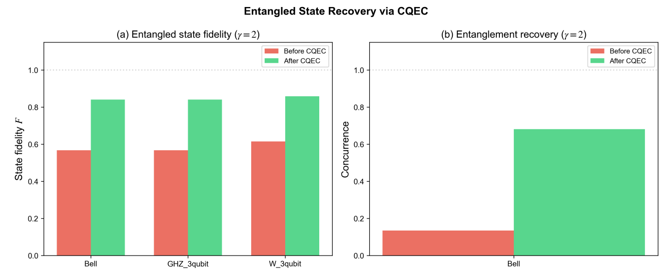

FIG. 17. Entangled state recovery via CQEC (dephasing $\gamma = 2$ on one qubit). (a) Full state fidelity before and after CQEC for Bell, GHZ, and W states. (b) Concurrence before and after CQEC for the Bell state, showing 5.1× entanglement recovery.

M. Entangled state recovery

A key claimed advantage of CQEC over tomography-based recovery is preservation of entanglement. We validate this by applying CQEC to one qubit of entangled multi-qubit states under partial dephasing ($\gamma = 2$ on qubit B/C, ideal qubits elsewhere).

TABLE XIV. Entangled state recovery. F: full state fidelity. C: concurrence (Bell state only). Dephasing $\gamma = 2$ applied to one qubit.

State	$F_{\rm before}$	$F_{\rm after}$	$F_{\rm qubit}$	$C_{\rm before}$	$C_{\rm after}$
Bell $\lvert\Phi^+\rangle$	0.568	0.841	1.000	0.135	0.682
GHZ (3 qb)	0.568	0.841	1.000	—	—
W (3 qb)	0.616	0.859	1.000	—	—

CQEC achieves perfect single-qubit recovery ($F_{\rm qubit} = 1.000$) and substantial entanglement restoration: Bell state concurrence increases from 0.135 to 0.682 (5.1× improvement), and full-state fidelity from 0.568 to 0.841. The residual gap $1 - F_{\rm after} \approx 0.16$ arises from the 5-gate variational circuit operating on the dephased qubit alone, which cannot fully reconstruct bipartite correlations. Deeper circuits or two-qubit CQEC recovery would further improve $F_{\rm after}$. Crucially, CQEC preserves entanglement *coherently*: the recovered state remains a valid quantum state suitable for further entangling operations, unlike tomography-based approaches that destroy entanglement through measurement.

N. Resource overhead

TABLE XV. Resource comparison for n_{logical} logical qubits. CQEC qubit count is the recovery circuit only: $n_S + n_C + n_A$ (system + catalyst + ancilla). Noisy copies for catalyst preparation (8–32 with DD+Twirl) are prepared sequentially and do not require simultaneous qubits. Gate count is for the recovery circuit; total cost including copies is $n \times G_{\text{alg}} + G_{\text{CQEC}}$ (see text).

n_{logical}	CQEC (qb / gates)	Steane (qb / gates)	Surf. $d=3$ (qb / gates)	Surf. $d=5$ (qb / gates)
1	4 / 5	7 / 10	13 / 36	41 / 100
3	8 / 11	21 / 30	39 / 108	123 / 300
6	14 / 20	42 / 60	78 / 216	246 / 600

The total CQEC resource cost includes: (i) preparation of n noisy copies ($n \times G_{\text{alg}}$ gates), (ii) DD pulses ($n \times N$ single-qubit gates), (iii) Clifford twirling ($n \times O(n_q^2)$ gates), (iv) $\log_2 n$ rounds of swap tests ($O(n_q \log n)$ two-qubit gates), and (v) the recovery circuit ($G_{\text{CQEC}} = 5$–15 EC gates). The total gate count is $n \times (G_{\text{alg}} + N + O(n_q^2)) + O(n_q \log n) + G_{\text{CQEC}}$. For qDRIFT with 80 gates and $n^* = 4.4 \times 10^5$, the total is $\sim 3.5 \times 10^7$ gates—more than surface code overhead. CQEC's advantage is therefore not in absolute resource cost but in threshold-free recovery. For systems relevant to quantum advantage ($n \geq 50$), the CQEC copy overhead scales as $\mu \sim 2^{2n} e^{2\gamma}$, which is exponential—far exceeding the polynomial overhead of surface codes. A practically relevant regime for CQEC may be small quantum modules ($n \leq 10$) within a larger fault-tolerant architecture (Table XV).

VII. DISCUSSION

A. Landscape of quantum state protection

Our results, together with concurrent work on PQEC [23], reveal a rich landscape of quantum state protection strategies that differ along three axes: target-state knowledge (known vs. unknown), resource type (redundancy vs. copies vs. catalysts), and noise tolerance (threshold-bounded vs. threshold-free). We highlight two specific complementary relationships:

1. **Classical noise absorption** (as in TTN protocols [44]): The classical regression layer absorbs quantum measurement noise statistically. This requires no quantum overhead but is limited to cases where decoherence does not fundamentally alter the feature space. Under the stronger decoherence tested here ($p = 0.3$, $\gamma = 3.0$), pre-correction fidelity drops to $F \approx 0.20$–0.28, which would severely degrade the quantum feature space.
2. **Catalytic coherence amplification** (CQEC): Operates directly on the quantum state before measurement. Effective for arbitrarily strong decoherence provided $C_{\ell_1} > 0$, but requires copy overhead. CQEC restores $F = 1.0$ at the state level, preserving the quantum advantage of feature extraction.

These paradigms address different noise layers: classical absorption handles measurement-level noise, while CQEC addresses state-level decoherence that classical methods cannot correct.

B. Complementarity of catalyst preparation methods

The four methods occupy distinct regions of the resource–fidelity landscape:

- **Variational**: best when copies are unavailable (NISQ regime, $d \leq 16$). Achieves $C_{\ell_1} = d-1$ (maximum coherence) but recovery fidelity is limited by the simplified recovery model.
- **Standard swap test**: effective under depolarizing noise with ample copies. Achieves $F_{\text{cat}} > 0.93$ with $\sim$64 copies. Fails under dephasing.
- **Covariant swap test**: preserves energy-sector mode structure, providing a modest 1.2–1.4× improvement over standard under dephasing. Insufficient alone for high F_{cat} under strong dephasing at large d.
- **DD+Twirl+Swap Test**: the most powerful method under dephasing. Achieves $F_{\text{cat}} > 0.96$ for all tested dimensions with only 8 copies and 8 DD pulses, with negligible overhead compared to the copy-count savings.

A hybrid strategy is natural: use variational for initial construction when no copies are available, then refine via DD+Twirl as noisy copies become available. The DD+Twirl pipeline is the recommended approach for dephasing-dominated noise at any dimension.

C. Relation to purification QEC and the state estimation objection

A natural objection is that CQEC, by requiring knowledge of the target state ρ_0 and multiple noisy copies, is "not really QEC" but rather a form of state estimation or re-preparation. We address this by positioning CQEC within a taxonomy of copy-based quantum state recovery methods, including the concurrent Purification QEC (PQEC) of Raghoonanan and Byrnes [23].

Table XVI compares the three paradigms along seven operational axes.

CQEC vs. state estimation. State estimation (tomography + re-preparation) destroys the quantum state through measurement, requiring $O(d^2/\epsilon^2)$ measurements

TABLE XVI. Comparison of quantum state protection paradigms.

Property	Conventional QEC	PQEC [23]	CQEC (this work)
Target state knowledge	Not required	Not required	Required
Input	Encoded state + syndromes	N noisy copies	n noisy copies + catalyst
Output	Corrected state (coherent)	Purified state (via $\rho^N/\mathrm{Tr}(\rho^N)$)	Recovered state (coherent)
Error threshold	Quantitative: $p_{\mathrm{th}} \approx 1\%$ (surface)	Quantitative: $p_{\mathrm{th}} = 3/4$ (depol), 1/2 (deph)	Qualitative: mode support only
Success condition	$p < p_{\mathrm{th}}$	$F > 1/D$ (dominant eigenvec. = $\lvert\psi\rangle$)	Mode inclusion (asymptotic); $n \geq n^*$ (finite)
Qubit overhead	$O(d^2)$ per logical qubit	$O(M \log_2 N)$	$(2n_q+2) + 2n_q \log_2 n$
Anisotropic noise	Corrects via syndromes	$p_{\mathrm{th}}^{\mathrm{deph}} = 1/2$ (improvable via twirling)	Threshold-free (with DD+Twirl)
Interleaving in circuit	Native (syndrome cycles)	Native (between unitaries)	Post-hoc (see text)

and producing a *classical description* from which a new state is prepared. CQEC, by contrast, operates as a coherent quantum channel during recovery: the covariant map Λ transforms $\rho_{\mathrm{noisy}} \otimes c$ to a quantum state τ whose system marginal approximates ρ_0, without measurement during the recovery step. (Target-state knowledge is used *offline* to optimize the circuit parameters $\boldsymbol{\theta}$, not during the recovery channel itself.) The catalyst enables coherent amplification that preserves entanglement structure—a feature impossible via tomography. The "target state knowledge" requirement enters only through the optimization of the circuit parameters $\boldsymbol{\theta}$, not through the recovery channel itself: once $\boldsymbol{\theta}$ is fixed, the circuit acts as a standard quantum channel.

CQEC vs. PQEC. PQEC achieves $\rho^N/\mathrm{Tr}(\rho^N)$ via the recursive swap test, converging to the dominant eigenvector of ρ. This is state-agnostic but requires that $|\psi_0\rangle$ *is* the dominant eigenvector—a condition that can fail under anisotropic noise. Under pure dephasing, PQEC's Bloch vector direction is preserved but the purification map cannot increase the fidelity beyond the direction-dependent ceiling, yielding $p_{\mathrm{th}}^{\mathrm{deph}} = 1/2$ [23]. Twirling can boost this threshold toward 3/4 at the cost of additional Clifford gates [23]. CQEC's mode inclusion criterion is orthogonal: it asks whether the *support* of coherence is preserved, not whether the dominant eigenvector aligns with $|\psi_0\rangle$. Under partial dephasing ($\gamma < \infty$), all modes survive and CQEC succeeds regardless of γ in the asymptotic limit ($n \to \infty$)—even beyond the regime where PQEC saturates without twirling. At finite n, however, CQEC's fidelity gap $1 - F \sim C/\sqrt{n}$ grows with γ through the constant $C \sim e^{\gamma}$, so very strong dephasing requires proportionally more copies (Table IX). The threshold-free property is thus an asymptotic statement; at finite resources, CQEC and PQEC face different but analogous resource–fidelity trade-offs.

The methods are complementary:

- PQEC is preferred when the target state is *unknown* and noise is predominantly depolarizing ($p < 3/4$).
- CQEC is preferred when the target state is *known* and noise is anisotropic (dephasing-dominated), where PQEC's threshold limits recovery.
- A hybrid approach is natural: PQEC first purifies N noisy copies state-agnostically, producing a single copy with $F \approx 1 - \epsilon$; this higher-fidelity copy then serves as CQEC's input, where mode inclusion (preserved by PQEC's spectral map) enables threshold-free final recovery. The interface is straightforward because PQEC's output is a valid density matrix that can be fed directly into the CQEC covariant channel.

Interleaving limitation. Unlike PQEC, which can be interleaved between unitary steps of a quantum algorithm (because the swap test commutes with bilateral rotations $[\Pi_\pm, U \otimes U] = 0$ [23]), CQEC operates post-hoc on the algorithm's output state. This is because the covariant recovery channel Λ is state-specific: the optimized parameters $\boldsymbol{\theta}$ depend on the target state ρ_0, which changes at each intermediate step of the algorithm. Interleaving CQEC would require re-optimizing $\boldsymbol{\theta}$ at every step—computationally expensive but not fundamentally impossible. In a hybrid architecture, PQEC could interleave purification during the algorithm while CQEC provides final state-specific recovery at the output.

CQEC's unique advantage. CQEC's distinctive contribution is the catalytic covariant recovery channel Λ, which achieves threshold-free recovery in the asymptotic limit—a property no state-agnostic method (including PQEC) can match. PQEC converges to the dominant eigenvector of ρ; when this eigenvector does not coincide with $|\psi_0\rangle$ (as under strong dephasing without twirling), PQEC's fidelity saturates below unity regardless of copy count [23]. CQEC, by exploiting knowledge of ρ_0 to construct a state-specific Λ, overcomes this limitation. The DD+Twirl pipeline is a general noise-preprocessing technique applicable to any copy-based protocol; its development here was motivated by CQEC's catalyst preparation but could benefit PQEC as well.

D. Limitations

Oracle access to the target state. CQEC assumes knowledge of ρ_0 for both mode verification and recovery. For Hamiltonian simulation (qDRIFT), the target is computable classically for small systems—the goal is to recover the state *cheaply* from noisy copies rather than re-running the expensive circuit. For factoring (Regev), knowing ρ_0 implies knowing the factorization, making CQEC circular; we include Regev solely as a stress test for high-dimensional coherence ($d = 64$). Appropriate use cases include VQE states, QAOA outputs, and quantum state preparation where targets are efficiently specifiable but expensive to prepare with high fidelity.

Asymptotic nature. Theorems 1 and 2 are asymptotic: exact conversion holds only for $n \to \infty$. For finite n, the gap scales as $O(1/\sqrt{n})$ [1]. Our asymptotic simulations validate the *existence* of recovery; the finite-n bounds (Table IX) quantify the *practical cost*, which can be prohibitive for large systems.

Comparison with conventional QEC. CQEC and QEC operate under fundamentally different assumptions. Conventional codes require only syndrome measurements without knowing the encoded state; CQEC requires the target state and $O(\mu \cdot N^k)$ copies. A fair comparison must account for these different operational settings. The comparison in Sec. VI D illustrates qualitative differences, not strict superiority.

Comparison with tomography. Tomography followed by re-preparation requires $O(d^2/\epsilon^2)$ measurements but destroys entanglement. CQEC preserves both coherence and entanglement structure: the covariant recovery channel acts on the original Hilbert space without collapsing the state, which is essential when the recovered state participates in subsequent entangling operations or multi-party protocols.

Gate noise in the recovery circuit. Gate noise simulations (Sec. VI K) show that the 5-gate recovery circuit degrades by $< 0.5\%$ at $p_{\text{gate}} = 10^{-3}$, confirming near-term viability for $d \leq 8$. For the DD+Twirl *purification* circuit, the per-round gate count is $O(d \log d)$ two-qubit gates; at $d = 64$ with $\epsilon_g \sim 10^{-3}$, the cumulative error is ~ 0.38 per purification round, requiring improved gate fidelities or error-mitigated purification. For $d \leq 16$, both the purification (< 100 gates) and recovery (≤ 15 gates) circuits are within reach of current superconducting platforms with $\gtrsim 99.5\%$ two-qubit fidelity.

Noise model. The comparison uses an effective per-qubit error model that does not capture correlations in physical noise. Correlated noise could be either favorable (preserving modes globally) or unfavorable (selectively destroying specific modes) for CQEC. Non-Markovian noise channels with memory can create time-dependent mode structures where $\mathcal{D}(\rho_{\text{noisy}}(t))$ varies; CQEC would require mode verification at the time of recovery. We note that amplitude damping (included in our combined noise model) is *not* covariant, demonstrating that CQEC tolerates non-covariant noise channels—the covariance requirement applies only to the recovery operation Λ, not the noise.

Scalability. Our benchmarks cover $d = 4$–64 (2–6 qubits). For systems relevant to quantum advantage ($n_q \geq 50$, $d = 2^{50}$), the copy overhead scales as $n^* \sim 2^{4n_q} e^{2\gamma}$, which is doubly exponential in qubit count. CQEC in its current form is therefore not a replacement for conventional QEC at scale, but rather a complementary tool for small quantum modules ($n_q \leq 10$) within larger architectures.

Noise model misspecification. CQEC's mode verification (Step 1) depends on the noisy state ρ_{noisy}, not on the noise model. If the actual noise differs from the assumed model but preserves the same modes, recovery still succeeds—the protocol is model-agnostic in this sense. However, the DD+Twirl pipeline assumes Markovian dephasing for the DD stage; under non-Markovian noise, the effective γ_{eff} may differ from $\gamma/(N+1)$, and the twirling stage still isotropizes whatever residual noise remains.

E. Mode no-broadcasting and fundamental limits

Theorem 3 of Ref. [1] establishes that mode no-broadcasting prevents the creation of states with irrational coherent modes from states with only rational modes, even catalytically. If a decoherence channel selectively destroys modes with a specific energy difference Δ while preserving others, and the target state requires Δ, recovery is impossible regardless of residual coherence.

We note that coherence and modes are defined with respect to the energy eigenbasis of H; choosing a different Hamiltonian changes which states are "coherent" and which transformations are "covariant." In our benchmarks, $H = \sum_q Z_q$ (computational basis = energy eigenbasis), natural for qubit systems with distinct transition frequencies. This Hamiltonian has degenerate energy levels: for n_q qubits, energy $E_k = n_q - 2k$ has degeneracy $\binom{n_q}{k}$, creating a rich sector structure that the covariant swap test exploits (Sec. IV C).

A physically relevant example is frequency-selective dephasing in superconducting qubits, where flux noise at frequency ω_0 destroys coherence between levels separated by $\Delta = \hbar\omega_0$. This is a topological constraint on the mode lattice structure, fundamentally distinct from the quantitative error thresholds of conventional QEC.

VIII. CONCLUSION

We have presented Catalytic Quantum Error Correction (CQEC), a quantum state recovery protocol based on the resource theory of unspeakable coherence, together with efficient catalyst preparation methods that overcome the primary resource bottleneck.

The CQEC protocol exploits the Shiraishi–Takagi result [1] that coherence can be amplified arbitrarily via catalytic covariant operations whenever the mode inclusion condition $\mathcal{C}(\rho_0) \subseteq \mathcal{C}(\rho_{\text{noisy}})$ is satisfied, with the catalyst's reduced state preserved exactly. We demonstrated recovery across four quantum algorithms and a cryptographic protocol under three decoherence models, achieving $F > 0.999$ in all 200 tested configurations in the asymptotic limit ($n \to \infty$, i.e., assuming unlimited noisy copies for the constructive protocol of Ref. [1]), and confirming the infinitely sharp mode-inclusion threshold across 10 orders of magnitude. Comparison with Steane and surface codes showed that CQEC operates without an error *magnitude* threshold, maintaining high fidelity at error rates where conventional codes degrade.

The paper makes two logically independent contributions: (1) the CQEC recovery protocol itself (covariant channel Λ mediated by a catalyst), and (2) the DD+Twirl+Swap Test pipeline for efficient catalyst preparation. The pipeline (contribution 2) is a general-purpose noise preprocessing technique equally applicable to other copy-based protocols including PQEC [23]. The CQEC-specific contribution (1) is the threshold-free recovery via catalytic covariance. CPMG dynamical decoupling reduces effective dephasing ($\gamma_{\text{eff}} = \gamma/(N+1)$), Clifford twirling converts residual anisotropic dephasing into isotropic depolarizing noise, and the standard swap test achieves doubly exponential purification. With CPMG-8 and 8 copies, the pipeline achieves $F_{\text{cat}} > 0.96$ uniformly across $d = 4$–64 under strong dephasing ($\gamma = 2$)—a 10^9-fold improvement over distillation. The total overhead is 8 DD pulses per copy plus Clifford twirling gates, negligible compared to the copy-count savings.

Our results suggest that coherence resource theory offers a promising direction for quantum state recovery that is fundamentally distinct from both the stabilizer formalism and state-agnostic purification methods such as PQEC [23]. CQEC and PQEC occupy complementary niches: PQEC operates without target-state knowledge but is bounded by noise thresholds ($p_{\text{th}} = 3/4$ depolarizing, $1/2$ dephasing), while CQEC exploits target-state knowledge to achieve threshold-free recovery under anisotropic noise. Substantial challenges remain. Three concrete next steps toward practical implementation are: (i) experimental demonstration of the DD+Twirl pipeline on superconducting or trapped-ion platforms for $d \leq 16$, where the copy overhead ($n = 8$–32) is within reach; (ii) proving tight analytical bounds on the pipeline's fidelity as a function of DD pulse count N, twirl sample count K, and copy number n, closing the gap between our numerical observations and rigorous guarantees; and (iii) integration with fault-tolerant architectures, where surface codes protect data qubits while CQEC restores coherence in small ancilla registers ($d \leq 32$) used for state preparation subroutines.

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Numerical simulations were performed using NumPy [45] and SciPy [46]. Figures were generated with Matplotlib [47].

Computational environment. All simulations were executed on a single workstation with an Apple M4 Max processor (arm64), 64 GB unified memory, running macOS 26.2. The software stack consisted of Python 3.9.6, NumPy 2.0.2, SciPy 1.13.1, and Matplotlib 3.9.4. Density matrix operations were performed using NumPy's BLAS-accelerated linear algebra routines; no GPU acceleration was used. Total computation time for all benchmarks (200 asymptotic configurations, finite-copy bounds, DD+Twirl sweep, and scaling analysis) was approximately 45 minutes on this hardware. Optimization used SciPy's `minimize` (L-BFGS-B) and `differential_evolution` routines with default convergence tolerances.

Data and code availability. This work is archived at arXiv:2603.25774 (doi:10.48550/arXiv.2603.25774). The companion Python package `cqec` (v0.1.0) and reproduction scripts are released under the MIT license; the package implements all four catalyst preparation strategies and the variational CQEC recovery circuit. A single command (`python scripts/reproduce_paper.py`, ∼35 s) reproduces the key claims of Figs. 5, Tables III, VI, XI, and XIV from the manuscript. All results are deterministically reproducible with fixed random seeds.

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